**K L UNIVERSITY**

**Department of Electronics and Communication Engineering**

# CAPSTONE PROJECT REVIEW PROGRESS RECORD

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| **Review Stage:** Review–1 | **Date of Review: 03 / 10 / 2026** |
| **Project Title:** | Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators |
| **Cohort / Batch:** | 2024 – 2028 (Batch B4) |
| **Project Guide:** | Dr. Arun Nimmala, Associate Professor, Dept. of ECE |

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## Brief Description of the Project

*Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)*

Deep Convolutional Neural Network (CNN) inference on resource-constrained edge hardware is dominated by Multiply-Accumulate (MAC) units, where 8×8 multipliers consume substantial silicon area and power (68 LUTs on Xilinx Zynq-7020). Approximate computing offers significant energy and area reduction by relaxing arithmetic precision. However, uniform approximation across an entire network causes severe accuracy degradation or yields negligible hardware savings. Existing layer-wise mapping techniques require multi-day genetic search (e.g., ALWANN requires 7.5 CPU-days) or costly GPU retraining loops. This project develops a lightweight, single-pass sensitivity scoring metric computed in just 0.2 seconds without retraining or iterative search. By coupling static weight variance with dynamic first-order loss gradients extracted across 256 calibration images, our framework accurately quantifies layer-wise error resilience. Error-critical early layers (conv1) are allocated exact multipliers, intermediate layers receive compensated designs, and noise-resilient deep layers (conv4, accounting for 48.5% of total MACs) are assigned aggressive truncated multipliers. The heterogeneous datapath is realized as a synthesizable AXI4-Stream accelerator IP core for physical deployment on the Xilinx Zynq-7020 (PYNQ-Z2) FPGA SoC, maximizing silicon efficiency while preserving baseline classification accuracy.

## Progress Achieved up to This Review

*Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.*

1. **RTL Multiplier Library Design & Characterization:** Designed synthesizable Verilog modules for four 8×8 unsigned multiplier architectures (Exact, Truncated-4, Compensated-4, and OR-based) using explicit partial-product arrays. Conducted exhaustive behavioral simulation of all 65,536 input vectors in Icarus Verilog, extracting Mean Error (ME), MAE, NMED, and maximum error bounds.
2. **Circuit-Level Dominance Proof:** Synthesized the multiplier library in Xilinx Vivado targeting the Zynq-7020 FPGA. Proved that Compensated multipliers strictly dominate Truncated designs—both consume identical silicon resources (56 LUTs, –17.6% area vs. 68 LUTs exact) because the synthesis engine folds the +2^(T-1) (+8) rounding constant into existing carry-in logic at zero extra LUT cost, while cutting arithmetic error by ~35% (ME = –4.25 vs –12.25).
3. **Baseline CNN Architecture & INT8 Quantization:** Trained a 4-convolutional layer CNN on Fashion-MNIST in TensorFlow/Keras, establishing an exact 32-bit floating-point baseline test accuracy of 84.69%. Designed an affine INT8 quantization pipeline featuring unsigned 8-bit activations [0, 255] and signed-magnitude weights matching the unsigned hardware multiplier datapath.
4. **Sensitivity Metric Formulation (S1–S5):** Formulated candidate sensitivity metrics. Implemented a single-pass automatic differentiation probe using TensorFlow GradientTape across 256 calibration images (0.2s runtime). Revealed that conv1 is overwhelmingly sensitive (S=1.000) due to primitive edge extraction, whereas conv4 is resilient (S=0.014).
5. **Preliminary Sweep & Noise Floor Diagnosis:** Executed the 12-cell layer-by-layer validation sweep using TRUNC=4. Discovered that the CNN's inherent resilience absorbed the mild TRUNC=4 noise (NMED = 0.000188), producing accuracy drops within ±0.1% to 0.3% (statistical noise floor) and yielding an initial NO-GO correlation ($|\rho| < 0.40$). Formulated the plan to scale to TRUNC=6 for signal amplification.

**Signature of Project Guide:** _____ **Date: 03 / 10 / 2026**